

# Indian Institute of Information Technology Kalyani

## End Semester Examination, Spring 2024 - 2025

**Subject:** Operating Systems

**Time:** 03 Hrs.

**Paper Code:** CSC401

**Full Marks:** 50

**Instructions:**

- [1]. Answer all questions and each answer should be in your own words, otherwise, you will be awarded zero mark. [2] All parts of a question should be answered consecutively. [3]. Answer the questions to the points. [4]. Answer script should be neat and clean. [5]. Do not write anything on the question paper. [6]. Number of pages in the question paper: 03. [7]. Write the keyword “END” at the end of your answer.

**Q1.** Answer the following questions.

**[6 + 4 = 10 Marks]**

- (a) The following processes are being scheduled using a preemptive, Round Robin scheduling algorithm. Each process is assigned a numerical priority, with a higher number indicating a higher relative priority. In addition to the processes listed below, the system also has an **idle task** (which consumes no CPU resources and is identified as  $P_{idle}$ ). This task has priority 0 and is scheduled whenever the system has no other available processes to run. The length of a time quantum is 10 units. If a process is preempted by a higher-priority process, the preempted process is placed at the end of the queue.

| Process | Priority | Burst time | Arrival |
|---------|----------|------------|---------|
| $P_1$   | 40       | 20         | 0       |
| $P_2$   | 30       | 25         | 25      |
| $P_3$   | 30       | 25         | 30      |
| $P_4$   | 35       | 15         | 60      |
| $P_5$   | 5        | 10         | 100     |
| $P_6$   | 10       | 10         | 105     |

- (i) Show the scheduling order of the processes using a Gantt chart.
- (ii) What is the turnaround time for each process?
- (iii) What is the waiting time for each process?
- (iv) What is the CPU utilization rate?
- (b) A computer system has three processes  $P_1$ ,  $P_2$ , and  $P_3$ . These processes belong to two groups named  $GP_1$  and  $GP_2$ . The group weightage is 0.5 for both groups. The process  $P_1$  and  $P_2$  belongs to  $GP_1$ , and  $P_3$  belong to  $GP_2$ . The base priority for  $P_1$ ,  $P_2$ , and  $P_3$  is 30, 40, and 50, respectively. Assume that all the processes are processor bound and are ready to run. The priorities are calculated after every second, and between two priority calculations, 60 clock ticks (interrupt) are generated. For this system, using the **Fair share scheduling** algorithm shows the tabular representation of the process execution sequence for four seconds of time.

**Q2.** Given the following statement for the Banker's Algorithm. There are five processes  $P_1$ ,  $P_2$ ,  $P_3$ ,  $P_4$  and three resource types  $R_1$ ,  $R_2$ , and  $R_3$ . Consider the following snapshot of a system: **[2 + 4 + 4 = 10]**

| Process | Current Allocation |       |       | Maximum Demand |       |       | Available |       |       |
|---------|--------------------|-------|-------|----------------|-------|-------|-----------|-------|-------|
|         | $R_1$              | $R_2$ | $R_3$ | $R_1$          | $R_2$ | $R_3$ | $R_1$     | $R_2$ | $R_3$ |
| $P_1$   | 1                  | 0     | 0     | 3              | 2     | 2     | 6         | 3     | 2     |
| $P_2$   | 6                  | 1     | 2     | 6              | 1     | 3     |           |       |       |
| $P_3$   | 2                  | 1     | 1     | 3              | 1     | 4     |           |       |       |
| $P_4$   | 0                  | 0     | 2     | 4              | 4     | 4     |           |       |       |

- (a) What is the content of the Need matrix?
- (b) Is the system in a safe state? Justify your answer.
- (c) If a request from process  $P_1$  arrives for (0 2 2), can the request be granted immediately and show all the calculations?

**Q3.** Answer the following questions.

[5 + 5 = 10 Marks]

- (a) In the code below, three processes are competing for six resources labeled  $A$  to  $F$  (Every resource has one instance only). Using a resource allocation graph, show the possibility of a deadlock in this implementation. Modify the order of some of the get requests to prevent the possibility of any deadlock. You cannot move requests across procedures, only change the order inside each procedure. Use a resource allocation graph to justify your answer.

|                                                                                                                                                                                |                                                                                                                                                                                |                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>void P0() {   while (true)   {     get(A);     get(B);     get(C);     // critical section     // use A, B, C     release(A);     release(B);     release(C);   } }</pre> | <pre>void P1() {   while (true)   {     get(D);     get(E);     get(B);     // critical section     // use D, E, B     release(D);     release(E);     release(B);   } }</pre> | <pre>void P2() {   while (true)   {     get(C);     get(F);     get(D);     // critical section     // use C, F, D     release(C);     release(F);     release(D);   } }</pre> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- (b) Suppose the following two processes, foo and bar, are executed concurrently and share the semaphore variables  $S$  and  $R$  (each initialized to 1) and the integer variable  $x$  (initialized to 0). Can the concurrent execution of these two processes result in one or both being blocked forever? If yes, give an execution sequence in which one or both are blocked forever. Can the concurrent execution of these two processes result in the indefinite postponement of one of them? If yes, give an execution sequence in which one is indefinitely postponed.

|                                                                                                           |                                                                                                           |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <pre>void foo do {   semWait(S);   semWait(R);   x++;   semSignal(S);   semSignal(R); } while(1); }</pre> | <pre>void bar do {   semWait(R);   semWait(S);   x++;   semSignal(S);   semSignal(R); } while(1); }</pre> |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|

**Q4.** Answer the following questions.

[4 + 4 + 2 = 10 Marks]

- (a) Consider a computer system with 12KB of main memory and the size of the each frame/page is of 4KB. Assume that a process of size 20KB will be executed in the main memory with the following page reference string 0 1 2 3 4 1 2 0 1 2 1 4 1. Compute the page fault rate for the second-chance (clock) page replacement algorithm.
- (b) Consider a computer system with 12KB of main memory and the size of the each frame/page is of 4KB. Assume that a process of size 28KB will be executed in the main memory with the following page reference string and the corresponding modify bit.

|                  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Reference String | 0 | 1 | 3 | 6 | 2 | 4 | 5 | 2 | 5 | 0 | 2 | 3 | 1 | 2 | 5 | 4 | 2 |
| Modify bit       | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 |

Compute the page fault rate for the enhanced second-chance page replacement algorithm.

- (c) Assume that a computer system is using the two-level paging scheme. The size of the page is 4KB. Assume that the size of each entry in page tables is 4 bytes. The computer system used a 64-bit logical address and a 64-bit physical address. The first-level page table has  $2^{32}$  entries, and each second-level page table also has  $2^{20}$  entries. The system uses a single MMU register for the first-level page table entry. What is the size of the total memory consumed by the first-level and second-level page tables?

**Q5.** Answer the following questions.

[4 + 3 + 3 = 10 Marks]

- (a) Assume that a computer system executes a process in the main memory with the following page reference strings: 3, 3, 4, 2, 3, 5, 3, 5, 1, 4. The initial working set window size ( $\Delta$ ) is 4. The initial working set window at time  $t = 0$  contains the pages {5, 4, 1}, where page 5 was referenced at time  $t = -2$ , page 4 at time  $t = -1$ , and page 1 at time  $t = 0$ . What is the total number of page faults and the average number of page frames used by computing the working set at each reference?
- (b) Assume that a computer system follows “*fixed variable multiple partitions*” memory allocation method. The memory is partitioned as follows (in order):

| Block Number | Block Size |
|--------------|------------|
| 1            | 200KB      |
| 2            | 350KB      |
| 3            | 100KB      |
| 4            | 400KB      |
| 5            | 375KB      |

The Best-fit memory management algorithm places the following processes of size 218KB, 410KB, 115KB, 345KB, and 100KB (in order). What is the amount of internal and external fragmentations?

- (c) Suppose that a disk drive has 500 cylinders numbered 0 to 499. The drive currently serves a request at cylinder 143; the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is 86, 170, 213, 474, 148, 109, 222, 450, and 330. Starting from the current head position, what is the total distance (in cylinders) the disk arm moves to satisfy all the pending requests for the SSTF disk-scheduling algorithm?